main.c

Run ▶

```c
#include <stdio.h>
#include <string.h>

int main()
{
    char input[100], stack[100];
    int i = 0, j = 0;

    printf("Enter the input string (use i for id): ");
    scanf("%99s", input);

    stack[0] = '\0';

    printf("\nStack\tInput\tAction\n");

    while (input[i] != '\0')
    {
        stack[j++] = input[i++];
        stack[j] = '\0';

        printf("%s\t%s\tShift\n", stack, input + i);

        while (j > 0)
        {
            if (stack[j - 1] == 'i')
            {
                stack[j - 1] = 'E';
```

Output

```
Enter the input string (use i for id): i+i

Stack   Input   Action
i   +i  Shift
E   +i  Reduce E->id
E+  i   Shift
E+i     Shift
E+E     Reduce E->id
E       Reduce E->E+E

Input string is accepted.
```

main.c

Run ▶

```
            printf("%s\t%s\tReduce E->id\n",
                   stack, input + i);
        }
        else if (j >= 3 &&
                 stack[j - 3] == 'E' &&
                 (stack[j - 2] == '+' ||
                  stack[j - 2] == '*') &&
                 stack[j - 1] == 'E')
        {
            char op = stack[j - 2];
            j -= 2;
            stack[j - 1] = 'E';
            stack[j] = '\0';

            if (op == '+')
                printf("%s\t%s\tReduce E->E+E\n",
                       stack, input + i);
            else
                printf("%s\t%s\tReduce E->E*E\n",
                       stack, input + i);
        }
        else
            break;
    }
}

if (strcmp(stack, "E") == 0)
```

Output

```
Enter the input string (use i for id): i+i

Stack   Input   Action
i   +i  Shift
E   +i  Reduce E->id
E+  i   Shift
E+i     Shift
E+E     Reduce E->id
E       Reduce E->E+E

Input string is accepted.
```

main.c

Run ▶

```
                    stack[j - 1] == 'E')
            {
                char op = stack[j - 2];
                j -= 2;
                stack[j - 1] = 'E';
                stack[j] = '\0';

                if (op == '+')
                    printf("%s\t%s\tReduce E->E+E\n",
                           stack, input + i);
                else
                    printf("%s\t%s\tReduce E->E*E\n",
                           stack, input + i);
            }
            else
                break;
        }
    }

    if (strcmp(stack, "E") == 0)
        printf("\nInput string is accepted.\n");
    else
        printf("\nInput string is rejected.\n");

    return 0;
}

```

Output

```
Enter the input string (use i for id): i+i

Stack   Input   Action
i   +i  Shift
E   +i  Reduce E->id
E+  i   Shift
E+i     Shift
E+E     Reduce E->id
E       Reduce E->E+E

Input string is accepted.
```